

1.What is the key difference between publishing a dashboard on Tableau Public versus Tableau Server/ Cloud in terms of accessibility and data security?

- **Accessibility:** Dashboards on **Tableau Public** are accessible to anyone on the internet. On **Tableau Server/Cloud**, access is restricted only to authorized users within an organization.
- **Data Security:** **Tableau Public** offers no data security as the workbook and underlying data can be viewed and downloaded by anyone. **Tableau Server/Cloud** provides high security through authentication and permission controls.

2.When publishing a Tableau dashboard, what happens to a live connection compared to an extract? How does this affect data refresh and performance?

- **Live Connection:** Tableau maintains a direct link to the database. Every interaction triggers a new query, ensuring real-time data but potentially slower performance depending on the database speed.
- **Extract:** Tableau creates a snapshot of the data stored in its own memory. This significantly improves performance for complex visuals, but the data must be manually or automatically refreshed on a schedule to stay updated.

3.In Tableau Server or Cloud (conceptually), which feature allows administrators to control which users can view, interact with, or edit a published dashboard?

- Administrators use **Permissions** (often managed through Users, Groups, or Projects) to control specific capabilities such as viewing, interacting, or editing.

4.If you want users to view a dashboard but not download the underlying data, which permission setting would you adjust in Tableau Server?

- You would adjust the **"Download Full Data"** (or "Download Summary Data") permission setting by setting it to **Deny**.

5.Name two methods by which a Tableau dashboard can be embedded into a web page or website.

1. **Embed Code:** Using the HTML `<script>` or `<iframe>` tags provided by the "Share" button.
2. **Tableau Embedding API:** Using JavaScript libraries for a more integrated and interactive experience.

6. List two best practices you should follow when preparing a dashboard for public sharing to ensure data security and privacy.

1. **Data Anonymization/Aggregation:** Ensure all Personally Identifiable Information (PII) is removed or aggregated to prevent individual data points from being exposed.
2. **Disable Workbook Downloads:** In the Tableau Public settings, uncheck the option that allows viewers to download the workbook and its data.

7. After publishing a dashboard, users report a "Data not found" error. What is the most common cause related to data source settings?

- The most common cause is **Authentication failure**, often because the publisher did not select the **"Embed Password"** option when publishing, or the Server lacks the necessary **Database Drivers** to connect to the source.

8. When embedding a Tableau dashboard into a secure internal portal, what two additional security measures should be implemented to ensure only authorized users can access the embedded content?

1. **Single Sign-On (SSO) or Trusted Authentication:** To ensure users are automatically authenticated through the organization's secure network.
2. **Row-Level Security (RLS):** To ensure that users can only see the specific rows of data they are authorized to view based on their role or region.